

KING COUNTY HOUSING — EXPLORATORY DATA ANALYSIS

Finding the Right Home for Thomas Hansen

A data-driven recommendation on affordability, neighborhood, and timing

By Micheal Eshak

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What we'll cover today

01

Meet the client

Who am I ?
Who Thomas Hansen is and what he needs ?

02

Our data & approach

What we analyzed and how, in plain terms

03

Can he afford it?

Matching budget to family size

04

Which neighborhood?

Finding "nice" and family-friendly, within budget

05

When should he buy?

Does timing change the price?

06

Insights & recommendations

Our concrete advice for Thomas

Meet Thomas Hansen

Micheal Eshak

Data analyst

Who am I?

- I'm analyzing the King County housing dataset

THOMAS HANSEN

Home Buyer

Who is my client?

- A family of 7 — 2 parents and 5 kids
- Working with a tight budget ("no money")
- Wants a nice, family-friendly, socially connected neighborhood

THE TWO QUESTIONS HE NEEDS ANSWERED

- Where should he look? (Location)
- When should he buy? (Timing)

To answer both, we turned to the actual record of home sales across King County — not guesswork.

Our data & approach

21,596

home sales analyzed

70

neighborhoods (zip codes)
covered

13

months of sales data
(May 2014 – May 2015)

HOW WE GOT TO OUR ANSWERS

- Started with real King County home-sale records — prices, location, size, condition, and sale dates
- Cleaned the data: fixed an obvious data-entry error and handled a small number of missing values sensibly
- Translated Thomas's needs into three concrete, testable questions (next slides)
- Backed every answer with the actual sales numbers, not assumptions

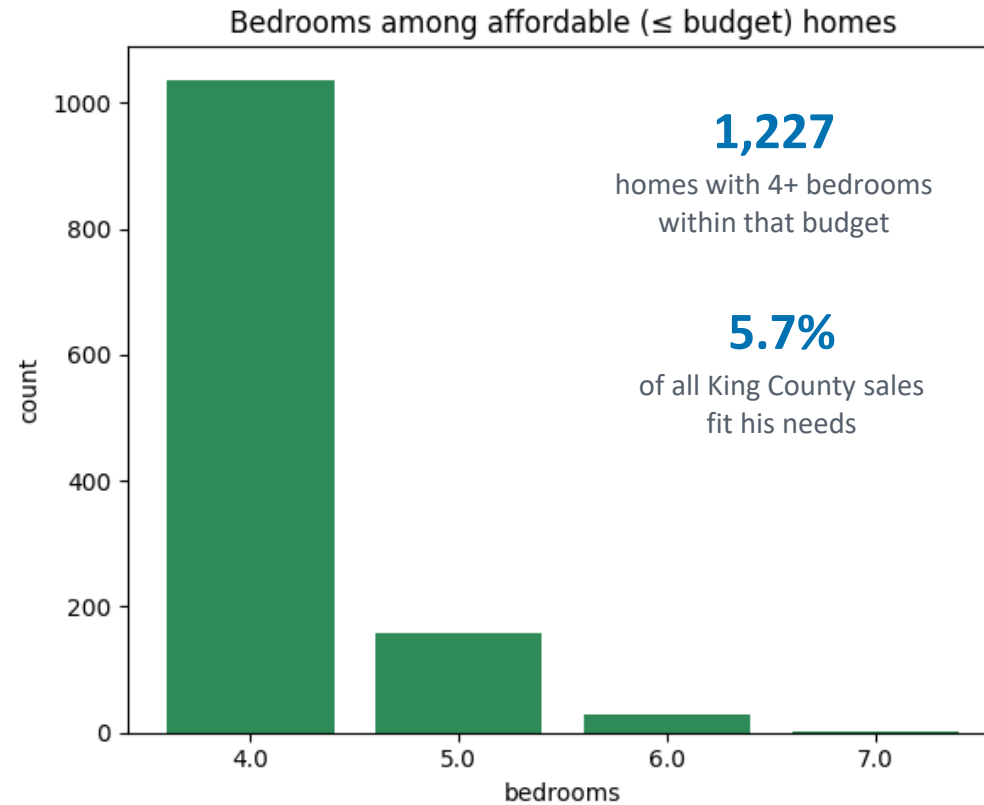
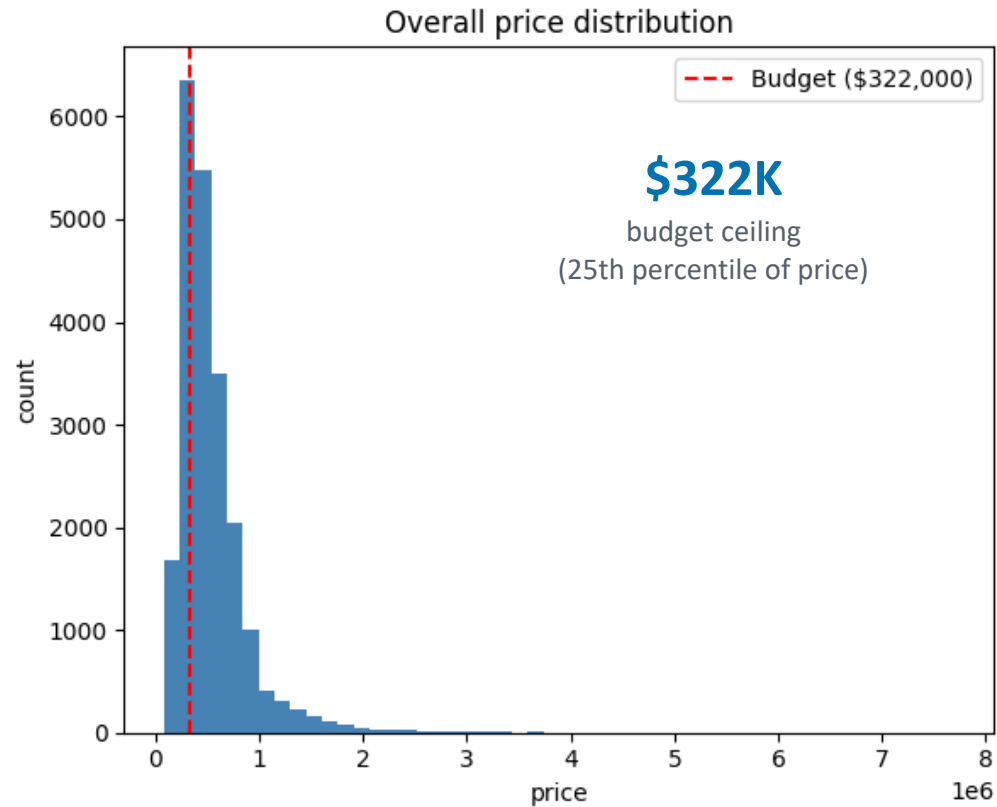
OUR THREE QUESTIONS

- Can Thomas afford a home big enough for his family?
- Which neighborhood is both “nice” and family-friendly, within his budget?
- Is there a better or worse time of year to buy?

QUESTION 1 — FEASIBILITY

Can Thomas afford a home big enough for his family?

We set his budget at the bottom 25% of all sale prices — a strict reading of “no money.”



Answer: Yes — a family of 7 is realistically housed within this budget in King County.

QUESTION 2 — LOCATION

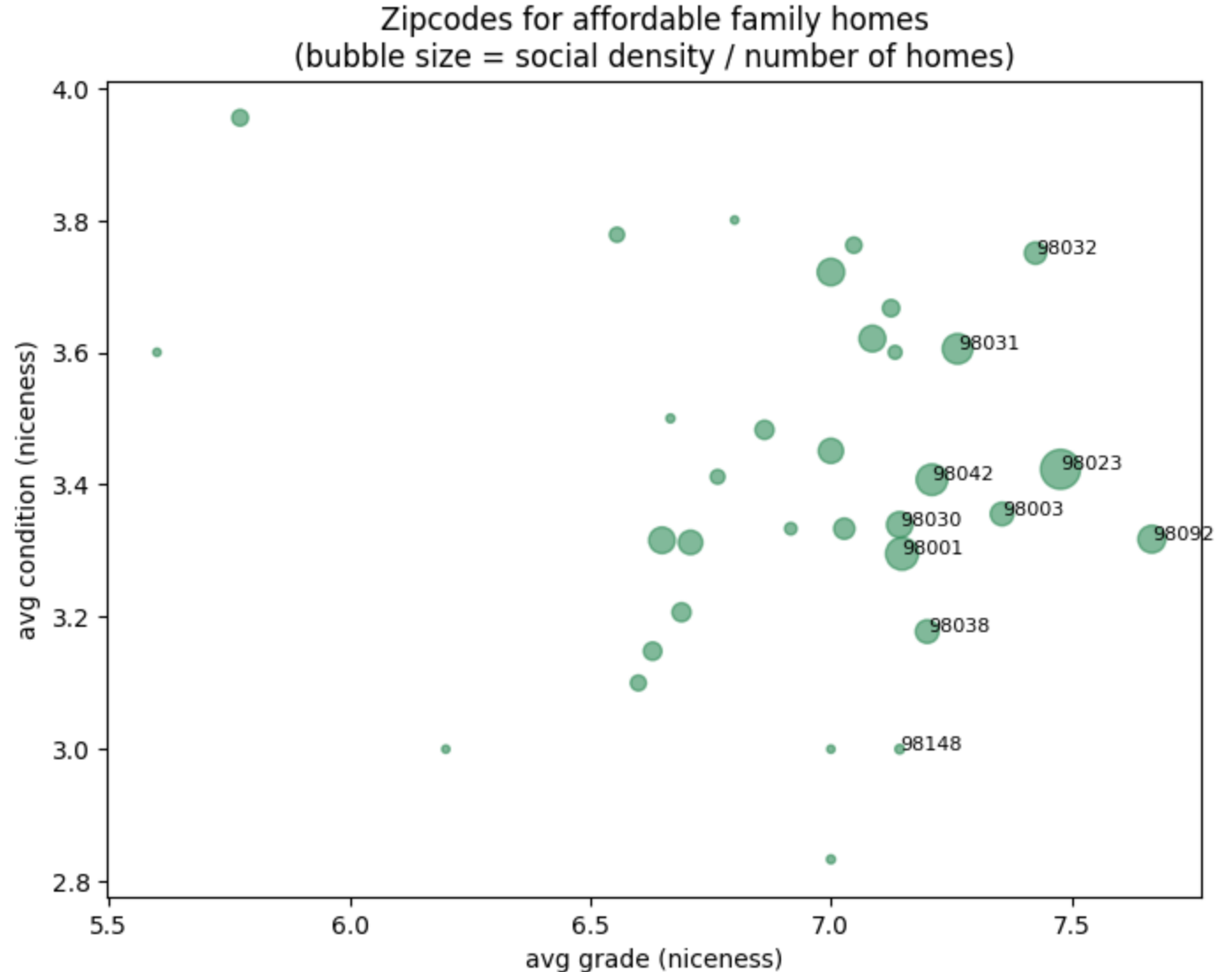
Which neighborhood is both “nice” and family-friendly?

Among the affordable, family-sized homes, we grouped by neighborhood and looked for areas that combine:

- “Social density” — lots of other similarly affordable, family-sized homes nearby
- “Niceness” — above-average home condition and build quality (grade)

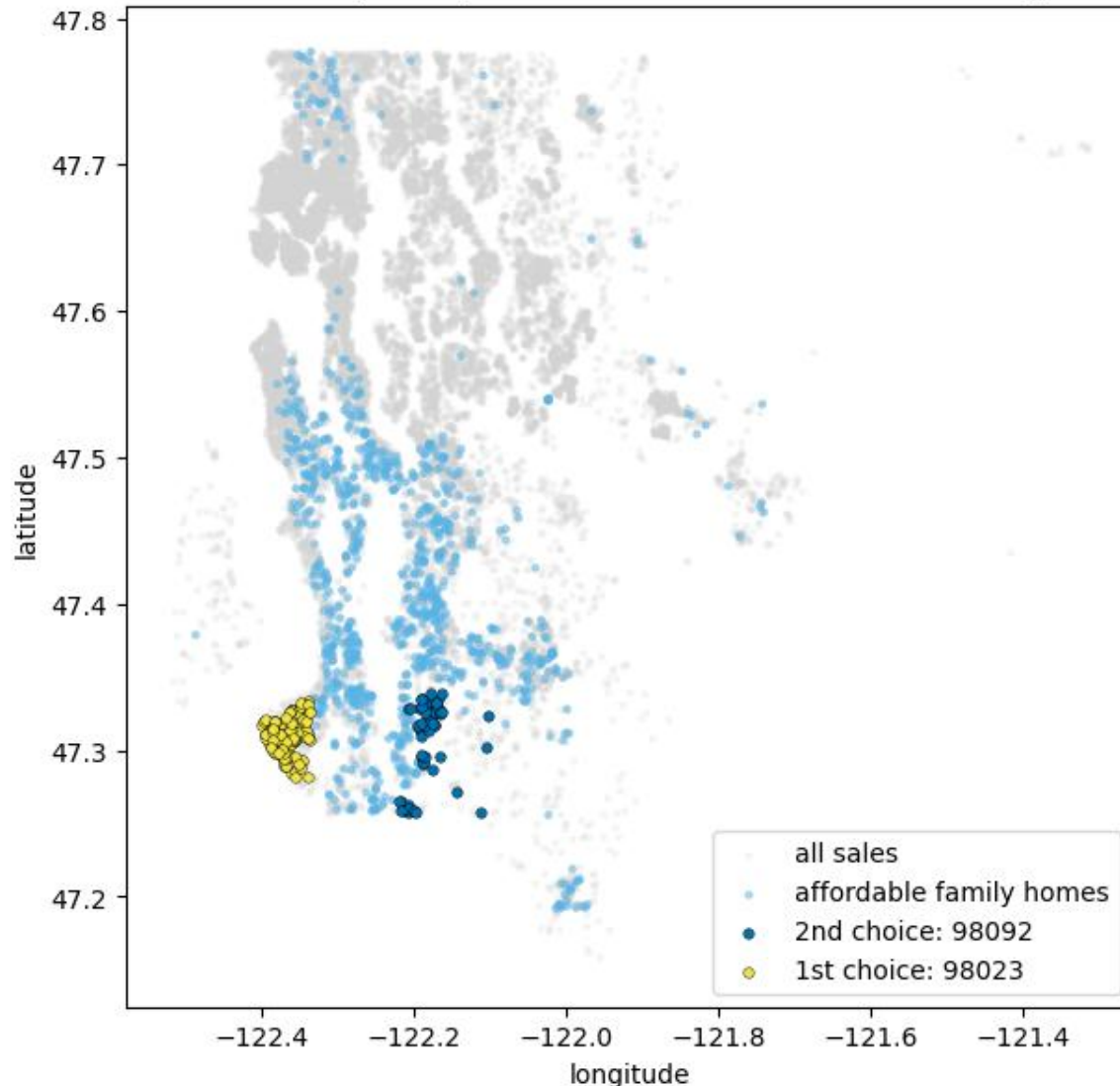
READING THE CHART

- Bigger bubble = more affordable family homes nearby
- Further right / up = better condition and build quality
- Two zip codes stand out: 98023 and 98092



Where these homes sit — and our top pick

Where affordable, family-sized homes are located across King County



★ 1ST CHOICE — ZIP CODE 98023

- 130 affordable, family-sized homes — by far the most of any neighborhood
- Lower average price: \$267,500
- Better average condition: 3.4 / 5
- Build quality (grade): 7.5 / 13

2ND CHOICE — ZIP CODE 98092 (backup)

- Highest average build quality (grade): 7.7 / 13
- But only 63 qualifying homes — far fewer choices
- Higher average price: \$281,000

QUESTION 3 — TIMING

Is there a better time of year to buy?

We tracked the average price of affordable, family-sized homes month by month across the year.

8%

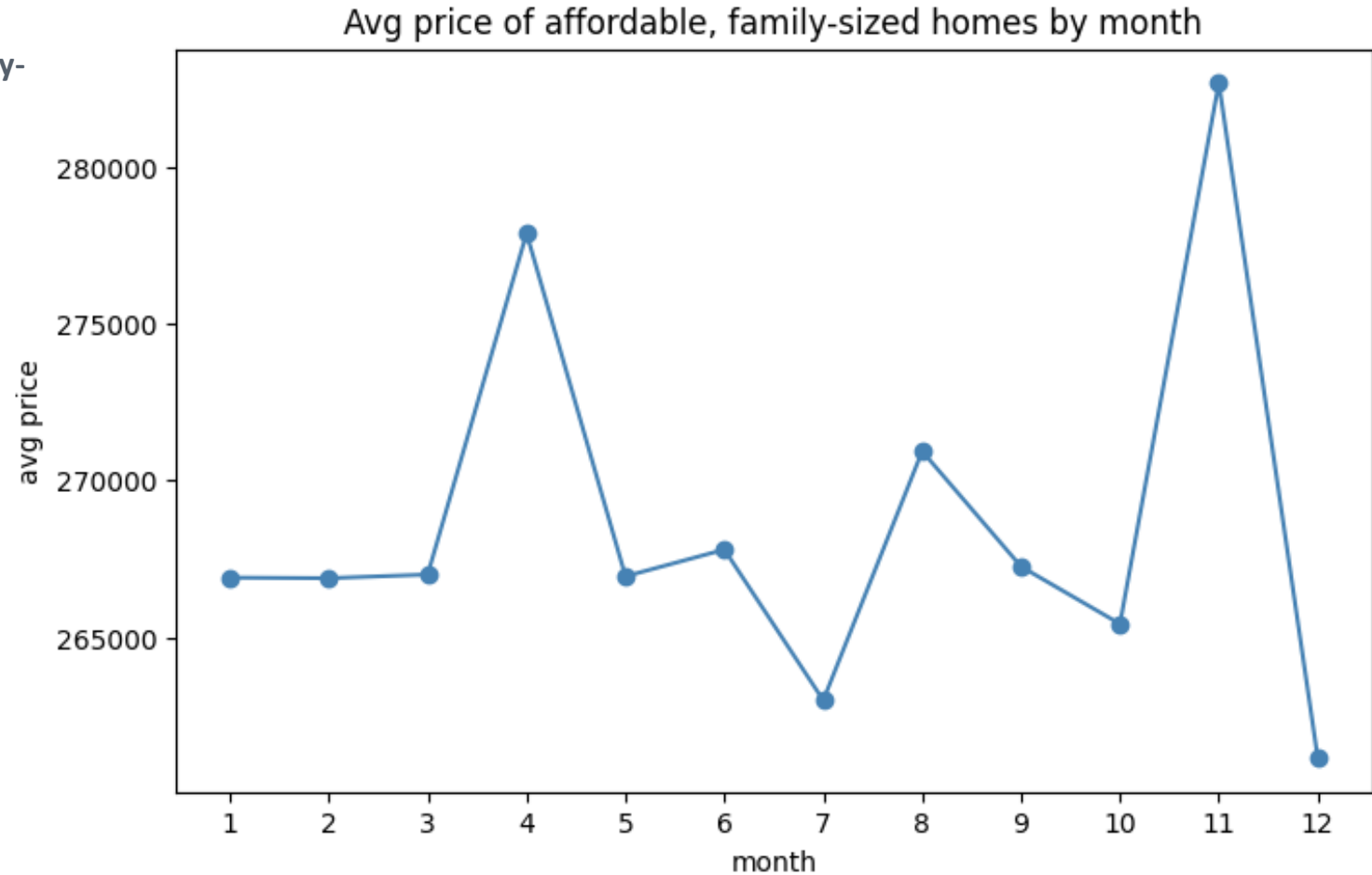
price swing across the
entire year — fairly flat

Dec

cheapest month
(avg. \$261,149)

Nov

priciest month
(avg. \$282,673)



Answer: Timing barely moves the needle — location matters far more than the calendar.

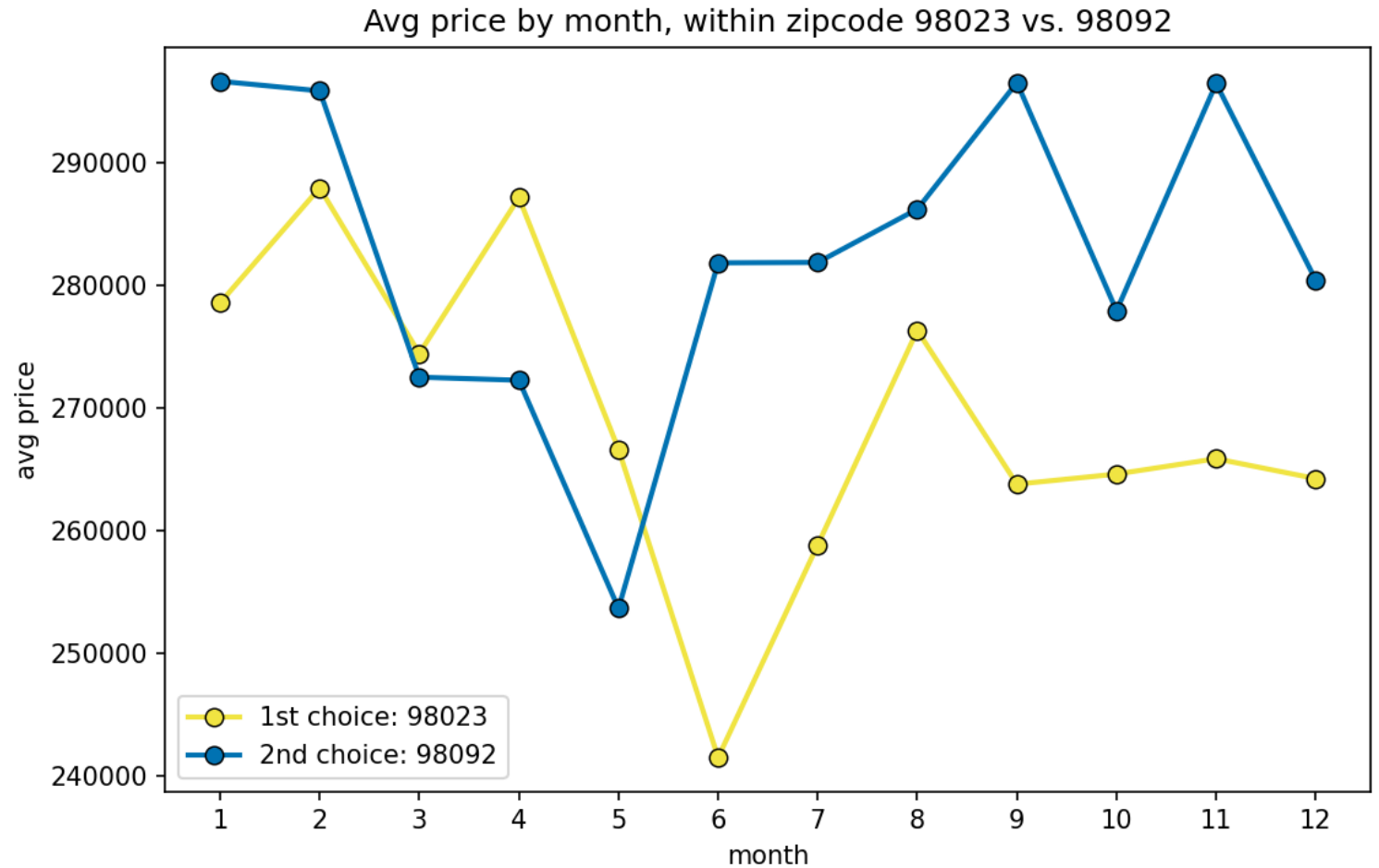
Does the pattern hold within each zipcode?

★ ZIP CODE 98023

- ~17% price swing across the year
- Cheapest month: June (avg \$241,512)
- Priciest month: February (avg \$287,922)

ZIP CODE 98092 (backup)

- ~15% price swing across the year
- Cheapest month: May (avg \$253,667)
- Priciest month: January (avg \$296,612)



Answer: Each zipcode has its own seasonal pattern — the countywide “December is cheapest” doesn’t apply to 98023 or 98092 specifically.

Key insights

1

Feasibility

Even on a strict budget, there are 1,227 homes with 4+ bedrooms available — a family of 7 is realistically housed in King County.

2

Location

Zip code 98023 stands out with the most affordable, family-sized homes of any neighborhood, at a price well under budget, with above-average condition and quality.
98092 is a strong backup.

3

Timing

The ~8% swing is a county-wide average — 98023 and 98092 actually move 15–17% individually.
Location, not the calendar, should still drive the decision.

OUR RECOMMENDATION

What we recommend for Thomas

1

Start the search in zip code 98023 first — it offers the best mix of affordability, family-home availability, and community density.

2

Keep zip code 98092 as a backup — slightly higher average build quality, but fewer homes on the market and a higher price.

3

Don't over-optimize for timing — within either zipcode, month-to-month swings (15–17%) are still smaller than what separates a home that fits his needs from one that doesn't.

Let location drive the decision.

4

If flexible on timing, use the zipcode-specific pattern, not the countywide one:
prices tend to be lowest around June in 98023, and around May in 98092 — not December.

Thank you

Questions & discussion